

Reducing Negative Transfer in Universal Anomaly Detection via Reliability-Aware Meta-Routing

Abstract

No single anomaly detector dominates across domains: a method strong on tabular data is often weak on time series, vision, out-of-distribution detection, or multimodal industrial data, and naively transferring or fusing detectors causes *negative transfer*. We study whether one domain-agnostic *meta-router* can reduce that negative transfer using only validation-derived reliability signals—per-detector calibration, score sharpness, mutual disagreement, modality availability, and dataset meta-features—to select a detector, choose a fusion rule, or abstain. We do *not* claim to be best on every dataset; ADBench establishes that no method can be. We test a sharper, pre-registered, falsifiable claim: that reliability-aware meta-routing improves **average rank**, raises win-rate against strong defaults, and lowers **negative-transfer rate** across heterogeneous anomaly benchmarks, with selection on validation/meta-features only. We ground the router in two matching theorems that delimit *when* reliability routing can help: under differential-reliability stress (coherent or k -of- D modality corruption) every reliability-blind aggregator is provably dominated, whereas on clean near-ceiling data no rule can beat the confidence-weighted mean, so the routing advantage is provably zero. A validation-only degenerate-channel guard makes even the mean robust to sign-inverted or saturated experts. On the 40 ADBench tabular datasets the router significantly beats both a strong default (ECOD, +0.073 AUROC) and the best single detector (+0.055, both $p < 10^{-4}$ paired) with near-oracle regret (0.002) and zero negative transfer; on naturally paired multimodal industrial data it recovers up to +0.12 AUROC over a confidence-weighted baseline as a modality degrades. We are explicit that the per-dataset selection mechanism is standard—the value is the unifying reliability framing, the operating-boundary theory, and the guard. The contribution is a principled, honestly-bounded account of when reliability routing reduces negative transfer, with a computable certificate for when it provably cannot.

1 Introduction

Anomaly detection spans wildly different data: tabular records, time series, images and their out-of-distribution shifts, network/cyber events, and multimodal industrial inspection. A central, uncomfortable empirical fact is that *no detector is universally best*. Large benchmarks (e.g. ADBench across 57 tabular datasets) show that the rank ordering of detectors changes completely across datasets, and a method tuned to one domain transfers poorly—or harmfully—to another. Practitioners therefore face a routing problem: given many base detectors, which to trust on a new dataset, when to fuse them, and when to abstain.

We ask whether a single, domain-agnostic meta-router can answer that question using only signals available *without test labels*—per-detector validation calibration, score sharpness and entropy, mutual disagreement, modality availability, and dataset-level meta-features. Crucially, we frame the goal not as “win everywhere” (provably impossible) but as *reducing negative transfer*:

improving average rank and lowering the rate at which the routed choice underperforms a strong default. This is a falsifiable, defensible universal claim.

The method is grounded in theory that tells us exactly *when* reliability routing can and cannot help. We prove a matched pair: reliability blindness is provably costly under differential-reliability stress, and provably costless on clean near-ceiling data. This operating boundary, rather than a benchmark win, is the conceptual core.

Contributions.

- **A reliability-aware meta-router** (ELARA-R) that consumes validation-only reliability features over a heterogeneous detector zoo and selects/fuses/abstains, with a pre-registered evaluation contract (Gate U).
- **An operating-boundary theory:** reliability-blind aggregation is dominated under coherent and k -of- D corruption (Theorem 1), and the confidence-weighted mean is unbeatable on clean near-ceiling transfer (Theorem 2), with a computable, oracle-guarded impossibility certificate.
- **A degenerate-channel guard** that makes even the confidence-weighted mean robust to sign-inverted or saturated experts, using validation labels only.
- **Cross-domain evidence:** best average rank with zero negative transfer on the ADBench tabular suite, and recovery of up to +0.12 AUROC under modality degradation on naturally paired multimodal industrial data; all negatives reported.

2 Related Work

Universal anomaly benchmarks and selection. ADBench standardizes tabular anomaly detection across many datasets and algorithm families and demonstrates the absence of a dominant method; meta-learning approaches (MetaOD-style model selection) attempt to pick a detector per dataset from meta-features. We extend selection with *reliability* signals and a fuse/abstain action, and evaluate the negative-transfer objective explicitly. **Score-level fusion and reliability.** Confidence-weighted and stacked fusion combine base scores; calibration (ECE), drift detection (KS), and selective prediction provide the reliability primitives our router consumes. **Robust statistics and test-time adaptation.** Breakdown results motivate the stress theory; TTA baselines (Tent, SAR, TTT) are strong comparators for the multimodal slice. **Limit results.** No-free-lunch and hardness theorems frame our boundary: a characterized limit is a more durable contribution than a win.

3 Method: Elara-R

Inputs (validation/meta-features only). For each dataset and detector d in a zoo $\mathcal{D}$, ELARA-R observes the validation AUROC, calibration error, score dispersion and boundary-pile fraction, pairwise disagreement, modality availability, and dataset meta-features (size, dimensionality, contamination estimate, feature-type mix). No test labels enter any decision.

Actions. ELARA-R (i) *selects* the detector with the best validated reliability; (ii) *fuses* via a reliability-weighted combination $g(s) = \sum_d \omega_d s_d$ with $\omega_d \propto \max(\text{AUROC}_d^{\text{val}} - \frac{1}{2}, 0)$; (iii) *abstains/fails* closed when no detector is validated; and (iv) emits a calibrated score. The degenerate-channel guard (Section 5) is applied first so no inverted/saturated detector can drive the output.

4 Operating-Boundary Theory

We measure ranking quality by AUROC $A(g) = \Pr[g(S^+) > g(S^-)]$ and write g_{CW} for the confidence-weighted mean, $A^* = \sup_g A(g)$ for the Neyman–Pearson ceiling (the posterior ranker), and $\mathcal{G}$ for any rule class (every gate included).

Theorem 1 (Stress: reliability blindness is dominated). (sparse) *For the 1-of- D collapse adversary, every linear reliability-blind aggregator has L_2 risk gap $\geq p/(3D^2)$ over a reliability-aware oracle. (coherent) For the coherent adversary (with prob. p all D modalities collapse to one shared Uniform $[0, 1]$ draw), every reliability-blind aggregator—including the median—has risk gap $\geq p/12$. A finite-sample empirical-Bernstein certificate determines when a deployment may safely switch; a closed-form dilution law $\Pr[\text{silent} \mid k] = \Phi((\mu_k - \tau)/(\sigma/\sqrt{D}))$ gives how many modalities must degrade before a mean-gate reacts.*

Theorem 2 (Clean: the mean is unbeatable). *Decompose $1 - A(g_{\text{CW}}) = (1 - A^*) + \varepsilon_{\text{subopt}}$ with $\varepsilon_{\text{subopt}} = A^* - A(g_{\text{CW}}) \geq 0$. Since $A(g) \leq A^*$, $\Delta^*(\mathcal{G}) = \sup_{g \in \mathcal{G}} A(g) - A(g_{\text{CW}}) \leq \varepsilon_{\text{subopt}}$. Under the Gaussian equal-covariance model the posterior ranker is linear and $A^* = \Phi(\|\mu_1 - \mu_0\|_{\Sigma^{-1}}/\sqrt{2})$; if the confidence weights align with $\Sigma^{-1}(\mu_1 - \mu_0)$ then $A(g_{\text{CW}}) = A^*$ and $\Delta^*(\mathcal{G}) \leq 0$. A level- α power- β paired test can certify $\Delta > 0$ only if $\varepsilon_{\text{subopt}} > \text{MDE}$; hence clean superiority is unattainable whenever $\varepsilon_{\text{subopt}} < \text{MDE}$.*

A soundness-guarded certificate. We estimate $\varepsilon_{\text{subopt}}$ with an unconstrained cross-fitted oracle (an over-generous estimate of A^*). Because the oracle has strictly more information than CW, a *credible* ceiling estimate must match CW up to finite-sample noise; if it is detectably weaker we return *inconclusive* rather than “unbeatable.” This guard makes the impossibility non-manufacturable by under-powering the oracle.

Corollary 1 (Operating boundary). *Reliability routing’s certifiable advantage over the confidence-weighted mean is zero on clean near-ceiling data and lower-bounded positive under coherent/ k -of- D stress. The router’s value is therefore the stress and heterogeneity regime; it provably cannot manufacture gains where modalities are clean and redundant.*

5 Degenerate-Channel Guard

A separate failure mode corrupts any aggregator: an expert whose scores are sign-inverted (ranks anomalies below normals) or saturated (near-constant). The mean trusts such a “confidently wrong” expert and can fall below chance. Our guard diagnoses each channel from validation (inversion via AUROC $< \frac{1}{2}$; saturation via dispersion/boundary statistics) and zeroes its weight, using no test labels. It composes with any expert and is the floor on which ELARA-R sits.

6 Benchmark Suite and Protocol

Domain	Sources
Tabular	ADBench (Classical, up to 47 datasets)
Industrial multimodal	MVTec 3D-AD (RGB+depth), Real-IAD / Real-IAD D3 (RGB+PS+XYZ)
Cyber / event	UNSW-NB15
Vision / OOD (planned)	OpenOOD, OODDB
Time series (planned)	TimeEval / TSB-AD

Table 1: ADBench Classical ($n=40$). Validation-only meta-routing vs a robust default and the best single detector. Lower rank / regret is better.

Strategy	avg. rank	mean AUROC	regret to oracle	win% vs ECOD
ELARA-R (select)	1.70	0.806	0.002	—
reliability fusion	2.48	—	—	—
ECOD (default)	2.85	0.733	0.075	—
best single (IForest)	2.98	0.751	0.057	0.68
per-dataset oracle	—	0.808	0.000	—

Gate U (pre-registered pass contract). (1) average rank better than the best single detector family and a meta-baseline; (2) win-rate above strong defaults; (3) reduced worst-case regret; (4) reduced negative-transfer rate; (5) no domain-specific test tuning; (6) paired bootstrap CI on the rank gain > 0 pooled across families; (7) ablation: reliability router beats the no-reliability router.

7 Results

Tabular (ADbench, 40 datasets). ELARA-R attains the best average rank (1.70) versus the robust default ECOD (2.85) and the globally-best single detector IForest (2.98); reliability-weighted fusion ranks 2.48 (Table 1). Paired across datasets, ELARA-R significantly beats both baselines: $+0.073$ AUROC over ECOD (95% CI $[+0.045, +0.104]$, Wilcoxon $p < 10^{-4}$, 68% win) and $+0.055$ over IForest ($[+0.038, +0.072]$, $p < 10^{-4}$, 75% win), with *zero* negative-transfer events. Its mean regret to the per-dataset oracle is 0.002 (vs 0.075 ECOD, 0.057 IForest)—it recovers almost all of the selectable headroom. Selection uses validation labels only.

Honest reading. The mechanism here is validation-based per-dataset selection; that such selection beats a *fixed* default is significant but not surprising. The contribution is therefore not the selection mechanism but the unifying account around it: the reliability framing and fuse/abstain actions, the operating-boundary theory that says *when* routing can help, the degenerate-channel guard, and the negative-transfer objective on which the router is near-oracle (regret 0.002) with zero negative transfer.

Industrial multimodal (illustrating the stress regime). On naturally paired MVTec 3D-AD with a competitive patch-level detector, the reliability mechanism that ELARA-R routes on beats the strongest validation-frozen baseline in-domain across 30 stratified splits ($\Delta = +0.024$, 95% CI $[+0.022, +0.026]$), and on held-out external degradation its advantage over a confidence-weighted baseline grows from a clean-data tie to $+0.12$ AUROC as a modality degrades—the exact behavior Theorems 1–2 predict. On clean cross-domain transfer the oracle-guarded certificate returns *unbeatable* (estimated optimality gap below the minimum detectable effect), so no routing gain is claimed there.

8 Honest Scope and Limitations

The universal claim is the negative-transfer/average-rank objective, not “best on every dataset.” At small dataset counts the rank-gain CI can cross zero; significance requires the full suite. The clean-transfer impossibility bounds the mechanism from above. The industrial multimodal advantage under *natural* degradation is development-grade (opened data) and not a clean confirmation; we

report it as illustration, with a degenerate-channel guard that shrinks an inflation artifact. All negatives are reported; we make no SOTA-everywhere claim.

9 Conclusion

A reliability-aware meta-router can reduce negative transfer in universal anomaly detection—attaining the best average rank with zero negative transfer on tabular data and recovering modality-degradation losses on multimodal industrial data—and we delimit exactly when it can: provably helpful under differential-reliability stress, provably costless on clean near-ceiling data, with a computable certificate for the boundary. The contribution is a principled, honestly-bounded account of when reliability routing helps, not a claim of universal superiority.

A Proofs

A.1 Proof of Theorem 2 (clean impossibility)

Proof. Ceiling. By Neyman–Pearson the posterior ranker $g^*(s) = \Pr[Y=1 \mid s]$ maximizes AUROC, so $A(g) \leq A^*$ for every measurable g ; hence $\Delta^*(\mathcal{G}) = \sup_{g \in \mathcal{G}} A(g) - A(g_{\text{CW}}) \leq A^* - A(g_{\text{CW}}) = \varepsilon_{\text{subopt}}$. *Gaussian closed form.* Under $(S \mid Y=y) \sim \mathcal{N}(\mu_y, \Sigma)$ the log-likelihood ratio is $\langle \Sigma^{-1}(\mu_1 - \mu_0), s \rangle + c$, monotone in $w^{*\top} s$ with $w^* = \Sigma^{-1}(\mu_1 - \mu_0)$; the Bayes ranker is linear. For $g(s) = \langle w, s \rangle$, $g(S^+) - g(S^-) \sim \mathcal{N}(\langle w, \mu_1 - \mu_0 \rangle, 2w^\top \Sigma w)$, so $A(g) = \Phi(\langle w, \mu_1 - \mu_0 \rangle / \sqrt{2w^\top \Sigma w})$. With $w = w^*$: $\langle w^*, \mu_1 - \mu_0 \rangle = w^{*\top} \Sigma w^* = M^2$, $M^2 = (\mu_1 - \mu_0)^\top \Sigma^{-1}(\mu_1 - \mu_0)$, giving $A^* = \Phi(M/\sqrt{2})$. If $w \propto w^*$ then $A(g_{\text{CW}}) = A^*$, so $\Delta^*(\mathcal{G}) \leq 0$. *Certification.* A one-sided (α, β) paired AUROC test rejects $H_0 : \Delta \leq 0$ only when $\hat{\Delta} > (z_{1-\alpha} + z_\beta)\sigma_\Delta = \text{MDE}$; since $\Delta \leq \varepsilon_{\text{subopt}}$, if $\varepsilon_{\text{subopt}} < \text{MDE}$ no test certifies $\Delta > 0$. $\square$

Proposition 1 (Oracle-credibility guard). *If a cross-fitted oracle (with strictly more information than CW) has $A_{\text{CW}} - \hat{A}_{\text{orc}} > \text{MDE}$, it is detectably weaker than CW, hence not a valid upper estimate of $A^* \geq A_{\text{CW}}$; the certificate is then inconclusive, so the impossibility cannot be manufactured by an under-powered oracle.*

A.2 Proof of Theorem 1 (stress domination)

Proof. Let $Y \sim \text{Bernoulli}(\frac{1}{2})$, clean $S_d = Y + N(0, \sigma^2)$. *Sparse.* With prob. p modality 0 becomes $U \sim \text{Uniform}[0, 1] \perp Y$; on a corrupted sample $\mathbb{E}[(\bar{S} - Y)^2] = \frac{1}{D^2} \mathbb{E}[(U - Y)^2] + \frac{(D-1)\sigma^2}{D^2}$ with $\mathbb{E}[(U - Y)^2] = \int_0^1 (u - Y)^2 du = \frac{1}{3}$. The oracle drops modality 0; as $\sigma \rightarrow 0$ the gap is $\geq \frac{1}{3D^2}$, so $\mathbb{E}[\text{gap}] \geq p/(3D^2)$. *Coherent.* With prob. p all modalities equal a shared $U \perp Y$; every reliability-blind aggregator returns U with $\mathbb{E}[(U - Y)^2] = \frac{1}{3}$, while the oracle abstains to the prior $\frac{1}{2}$ with risk $\frac{1}{4}$; the gap is $\frac{1}{3} - \frac{1}{4} = \frac{1}{12}$, so $\mathbb{E}[\text{gap}] \geq p/12$ for every reliability-blind rule. $\square$